

scripts/verify-codegraph.sh — user guide

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Overview

scripts/verify-codegraph.sh is the anti-bluff verification suite runner for the **codegraph** code-intelligence integration (see docs/CODEGRAPH.md and constitution §11.4.78). Per the Local-Only CI/CD constitutional constraint it **is** the codegraph quality gate — there is no hosted-CI equivalent.

It runs the six tests/codegraph/test_*.sh layers in order, tees per-layer evidence to .lava-ci-evidence/codegraph/<UTC-timestamp>/, writes a summary.md, and exits non-zero if any layer FAILs.

Prerequisites

- codegraph installed and on PATH (npm install -g @colbymchenry/codegraph).
- .codegraph/codegraph.db present — run codegraph index first if absent.
- python3 (used to drive the MCP JSON-RPC layers).
- For the full run (layer 05): the 5 CLI agents (claude, opencode, qwen, kimi, crush) installed; un-authenticated agents are reported as documented SKIP gaps, never as failures.

Usage

```
scripts/verify-codegraph.sh          # full run — all 6 layers,  
                                     incl. agent E2E  
scripts/verify-codegraph.sh --quick  # layers 01-04 + 06 only  
                                     (skip slow LLM layer 05)
```

Environment overrides:

- CODEGRAPH_EVIDENCE_DIR — override the evidence output directory.

- CODEGRAPH_AGENT_TIMEOUT — per-agent timeout in seconds for layer 05 (default 240).

The six layers

Layer	File	Proves
01	test_01_index_reality.sh	codegraph indexed the real Lava codebase; secrets/submodules excluded
02	test_02_query_correctness.sh	codegraph query resolves real symbols to real file:line
03	test_03_mcp_protocol.sh	the MCP server returns real data over real stdio JSON-RPC
04	test_04_agent_connectivity.sh	all 5 CLI agents register/connect to codegraph
05	test_05_agent_e2e.sh	each agent provably <i>uses</i> codegraph in a real LLM turn (unforgeable node-count challenge)
06	test_06_falsifiability.sh	the suite FAILs when codegraph is deliberately broken, PASSes when restored

Edge cases

- **--quick** skips the slow LLM-driven layer 05 — use it for fast local checks; the full run is required before relying on the integration.
- **Documented gaps (SKIP):** layer 05 marks an agent SKIP when it cannot be run in the environment (missing credentials, exhausted quota, an agent-environment incompatibility). A SKIP is a documented gap, never a faked PASS (§6.J / §6.L); it does not fail the suite, but is reported prominently.
- **Pre-flight failure (exit 2):** if codegraph is not installed or .codegraph/codegraph.db is missing, the runner aborts before running any layer.
- **Falsifiability rehearsal** (layer 06) moves .codegraph/codegraph.db aside and restores it; a trap restores the DB even if interrupted.

Internal behaviour

The runner sources nothing; each `tests/codegraph/test_*.sh` sources `tests/codegraph/lib.sh` for shared assertions and the MCP JSON-RPC drivers. Each layer exits 0 (all assertions pass) or 1 (any fail). The runner aggregates PASS/FAIL/SKIP counts from the per-layer logs and writes `summary.md`. Exit codes: 0 suite passed, 1 a layer FAILED, 2 pre-flight failed.

Related scripts

- `tests/codegraph/lib.sh` — shared library (assertions, MCP drivers).
- `tests/codegraph/test_0[1-6]_*.sh` — the six verification layers.
- `docs/CODEGRAPH.md` — the codegraph integration reference.

Last verified

2026-05-20 — full run: 45 pass · 0 fail · 4 documented SKIP gaps.